Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

# ENVIRONMENTAL SCIENCE

## Paper 2 (Generated practice paper - Set 06)

Assessed: The living environment, Biological resources, Sustainability, Research methods

**Time allowed: 3 hours**

### Materials

For this paper you may use:
- a calculator.

### Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

### Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| 7 | |
| 8 | |
| 9 | |
| 10 | |
| 11 | |
| TOTAL | |

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**Answer all questions in the spaces provided.**

**0 1 . 1** Describe how each of the following abiotic factors can be managed to increase biodiversity in a nature reserve.

**[6 marks]**

**Light** .....

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**Dissolved oxygen** .....

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**Nutrients** .....

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**0 1 . 2** Explain how biotic factors can be managed to increase the survival of a species. Give **four** examples.

**[4 marks]**

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**10**

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**0 2 . 1** Describe **four** ways in which the Wildlife and Countryside Act (1981) protects wildlife in the UK.

**[4 marks]**

**1** .....

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**2** .....

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**3** .....

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**4** .....

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**0 2 . 2** Explain the difference between CITES Appendix I and Appendix II, giving an example species for each.

**[4 marks]**

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**0 2 . 3** Explain why controlling international trade can be more effective than trying to stop collection of a species in its own country.

**[2 marks]**

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**10**

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**0 3 . 1** Explain why a moderate water shortage reduces crop productivity even though the plants survive.

**[3 marks]**

**0 3 . 2** Explain the problems caused by waterlogged soils and describe **two** ways farmers reduce soil water levels.

**[4 marks]**

**0 3 . 3** Explain how irrigating with groundwater can cause soil salinisation and how this harms crops.

**[2 marks]**

**0 3 . 4** State how a farmer could raise the pH of an acidic soil and explain why a low pH reduces fertility.

**[1 mark]**

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**0 4 . 1** Describe the main principles of the circular economy.

**[6 marks]**

**0 4 . 2** Explain what is meant by 'optimum production rather than maximum production', using a comparison between a natural ecosystem and an agroecosystem.

**[2 marks]**

**0 4 . 3** Explain why international cooperation (eg through the EU or UN) is needed to implement circular economy initiatives.

**[2 marks]**

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**0 5 . 1** Explain how natural material cycles differ from linear human systems of material use.

**[4 marks]**

**0 5 . 2** Explain why the use of toxic metals in electronic products is a particular problem for sustainability.

**[2 marks]**

**0 5 . 3** Describe how the design of a product can move it from a linear to a cyclical system.

**[4 marks]**

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**10**

**0 6 . 1** Which of the following features of a plantation would most increase its wildlife value? Tick one box.

**[1 mark]**

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

**0 6 . 2** Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

**[4 marks]**

**1** .....

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**2** .....

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**3** .....

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**4** .....

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**5**

**0 7 . 1** Describe **three** methods of reducing seabird by-catch in long-line fisheries.

**[3 marks]**

**1** .....

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**2** .....

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**3** .....

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**0 7 . 2** Explain how restricting fishing effort and restricting fishing methods reduce overfishing.

**[3 marks]**

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**0 7 . 3** Describe how captive rearing and release ('population seeding') can increase wild populations of lobsters or oysters.

**[2 marks]**

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**0 7 . 4** Explain why collecting data on planktonic fish eggs and larvae helps to prevent population collapse.

**[2 marks]**

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**0 8 . 1** Explain how overfishing of the giant triton mollusc damaged coral reefs in Australia.

**[2 marks]**

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**0 8 . 2** Describe **four** management practices used in marine protected areas to protect coral reefs.

**[4 marks]**

**1** .....

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**2** .....

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**3** .....

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**4** .....

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**0 8 . 3** Explain why deep-water coral reefs recover much more slowly from trawling damage than tropical reefs.

**[2 marks]**

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**0 8 . 4** Give **two** threats to deep-water reefs and **one** conservation measure.

**[2 marks]**

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**0 9 . 1** Explain how each of the following changes in abiotic factors caused by human activities can make a habitat unsuitable for species living there.

**[6 marks]**

**Water availability**

**Dissolved oxygen**

**pH**

**0 9 . 2** Explain how increased water turbidity caused by ploughing, mining or dredging harms aquatic organisms.

**[2 marks]**

**0 9 . 3** Explain why species that were already rare are most likely to be lost when abiotic conditions change.

**[2 marks]**

10

**1 0 . 1** Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

**[9 marks]**

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Extra space

**1 0 . 2** State what is meant by a **taxon**.

**[1 mark]**

**10**

**Write an essay on one of the following topics.**

**11 . 1** Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

**[25 marks]**

**OR**

**11 . 2** Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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**END OF QUESTIONS**